ENDZONE EQUITY

December 12, 2025

END ZONE EQUITY

FINAL REPORT

WInS Username: RamazThree-9684492

Advisor: Ms. Linda Lowry

Benjamin R. | **Team Leader / Head Coach**

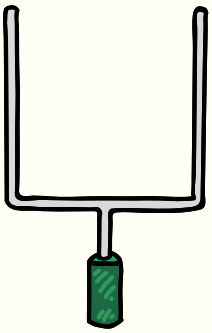
Adam S. | **Defensive Coordinator**

Caleb D. | **Special Teams Coach**

Noah S. | **Offensive Coordinator**

Samuel D. | **Playbook Operator**

Victoria D. | **Salary Cap Manager**



FINAL REPORT
Advisor and Team Reflections

Advisor Reflection

At Ramaz, participating in the Wharton High School Investment Competition generates excitement and anticipation as a club activity. The competition kicked off on September 8 with a general meeting on team structure, the trading process, competition time frame, and deliverables. Further general meetings were held to assess the client's risk profile and to discuss various investment strategies.

For End Zone Equity, a junior team, the majority of its members participated in the competition last year. Given this prior experience, they had a solid understanding of the fundamentals of investing. The team got off to a quick start by dividing up responsibilities such as large cap or small cap strategies. Ben, the team leader, did an excellent job coordinating their efforts which he discussed frequently with me. With another year of competition, the team enhanced their investment knowledge, allowing them more creativity in their approach to meeting the client's needs.

Team Reflection

As a team, we are uniquely positioned in life to access excellent financial literacy opportunities and resources that many teenagers don't have. We have an ethical responsibility to use our knowledge for the benefit of our community. Many people will never experience the education we have been given, nor the guidance that comes with it. Therefore, we want to make a meaningful contribution to the community. This means setting up financial literacy workshops and student-to-student mentorship programs, teaching underprivileged people how to think about their money responsibly. By giving them investment opportunities and economic curricula, we encourage long-term money planning in an educated and well-mannered way. This instills values like learning, communication, and financial literacy as pillars to empower the next generation of bright young thinkers. However, the true goal is to ensure that everyone has access to amazing opportunities.

FINAL REPORT
Trading Notes Analysis 1

Trading note #1 (Stock)

"HP Inc. is a part of the Large Cap portion of our portfolio. When comparing this stock to other stocks in its sector from the S&P 500, it scored in the top 30% with a score of 14.7 on our quantitative index. It also scored a 61.2 on our Scouting Report, which is above the 40th percentile." - HPQ

HP earned a quantitative index score of 14.7, ranking 4th of 64 IT stocks and giving it a 14.6% weight within the sector. With Information Technology projected to grow 7.9% over the next six months and representing 35.6% of our large-cap allocation (about 21% of the total portfolio) HP ultimately holds a 3.2% portfolio weight. HP advanced to our Scouting Report evaluation, scoring 114.9 for sustainability language and 25.4 for community-focused language in its 10-K report. Its weighted SR score of 61.2 ranked 4th of 19 in its sector.

Trading note #2 (ETF)

"VEA is a part of the ETF portion of our portfolio. VEA is an international fund that allows us to diversify into well-established, large-cap companies in overseas markets. It has a strong yield of 2.80%."

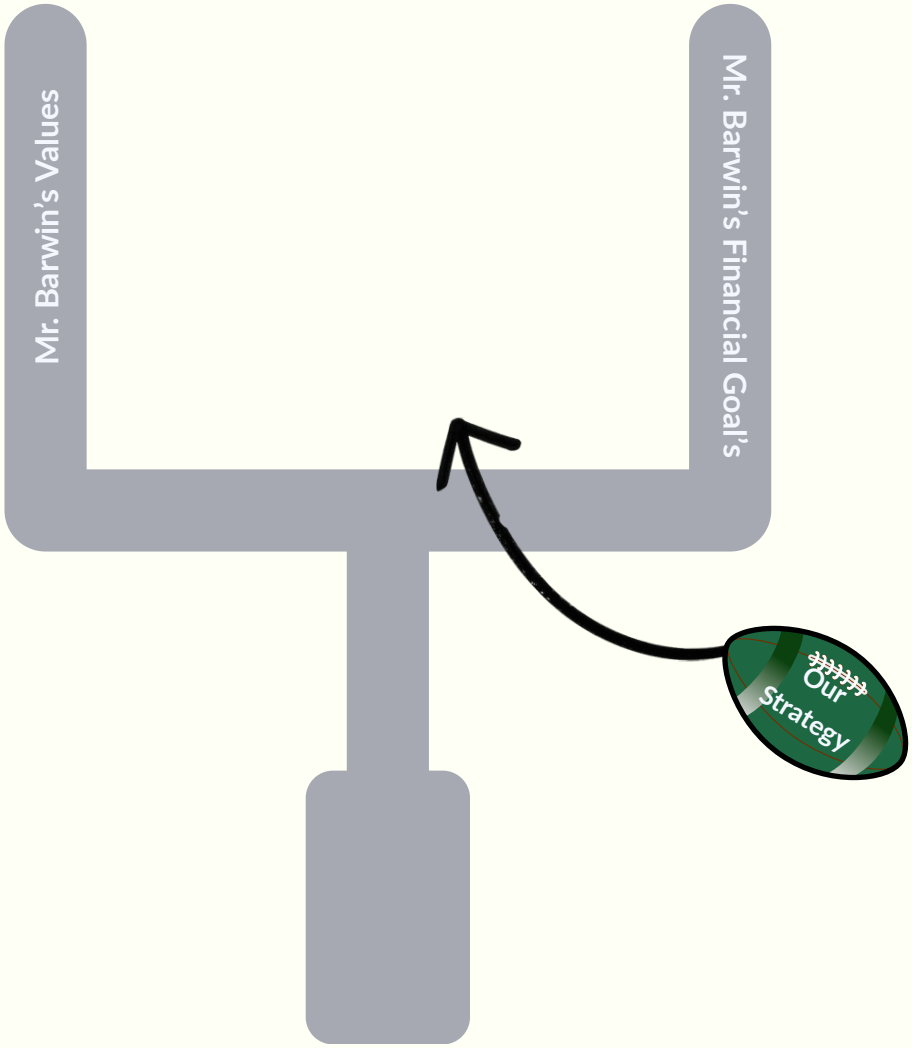
ETFs serve primarily as a diversification tool in our portfolio, and international exposure is a critical component of that diversification. VEA – being a large-cap international-based ETF – is a reliable, growth-oriented investment capable of providing stability if domestic markets soften. As mentioned in the note, VEA has a high, 2.8% yield, which is essential to us because it strengthens the annual withdrawal fund for Mr. Barwin, which we intend to support through the portfolio's dividend income.

Trading note #3 (Stock)

"OSS is part of the small-cap portion of our portfolio. It passed the TAM analysis and scored a 34 on our quantitative index."

OSS is part of the small-cap portion of our portfolio. It passed the TAM analysis and scored a 34 on our quantitative index. The TAM analysis confirmed that OSS operates in a large and expanding AI and edge computing market, which explains their continued reinvestment despite weaker operating cash flow. Its quantitative score reflected solid revenue expansion, which we validated through strong quarterly revenue growth. OSS helped shape our strategy by showing how quantitative scoring and TAM analysis must work together to help distinguish between strategic reinvesting and unsustainable cash burn in small-cap investing.

OSS
ONE STOP SYSTEMS



ENDZONE EQUITY

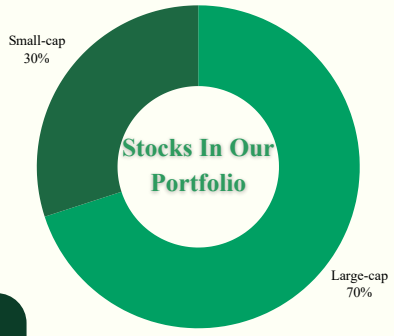
FINAL REPORT
Portfolio at a Glance 1

STOCKS

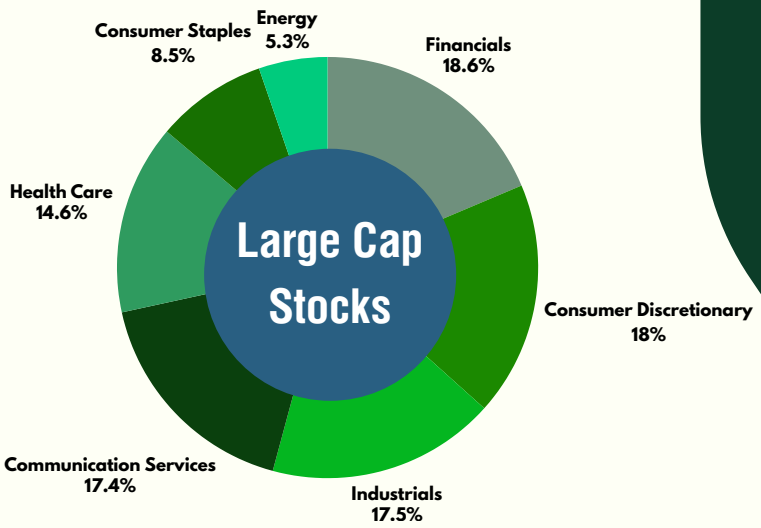
Stock Ticker	Company Name	Value ($)	% of total portfolio
ADM	Archer-Daniels-Midland Co	$1,750	0.35%
ALLE	Allegion PLC	$1,100	0.22%
AMAT	Applied Materials Inc	$5,350	1.07%
AMD	Advanced Micro Devices Inc	$1,950	0.39%
APH	Amphenol Corp	$5,000	1.00%
AXP	American Express Co	$2,250	0.45%
BAC	Bank of America Corp	$1,600	0.32%
BG	Bunge Global SA	$5,350	1.07%
BKR	Baker Hughes Co	$2,750	0.55%
BLDR	Builders FirstSource, Inc.	$750	0.15%
C	Citigroup Inc	$4,700	0.94%
CAH	Cardinal Health Inc	$2,800	0.56%
CAT	Caterpillar Inc	$3,000	0.60%
CCL	Carnival Corp	$1,950	0.39%
CFG	Citizens Financial Group Inc	$2,550	0.51%
CMCSA	Comcast Corp	$6,250	1.25%
CMI	Cummins Inc	$2,550	0.51%
COR	Cencora Inc	$2,050	0.41%
DAL	Delta Air Lines Inc	$4,850	0.97%
DHI	DR Horton Inc	$2,850	0.57%
DOV	Dover Corp	$2,100	0.42%
DVN	Devon Energy Corp	$1,750	0.35%
EBAY	eBay Inc	$2,300	0.46%
EQT	EQT Corp	$1,650	0.33%
EXPD	Expeditors International of Washngtn Inc	$1,950	0.39%
F	Ford Motor Co	$5,100	1.02%
FDX	FedEx Corp	$1,350	0.27%
FOXA	Fox Corp Class A	$2,750	0.55%
TWST	Twist Bioscience Corp	$6,800	1.36%
OSS	One Stop Systems Inc	$9,300	1.86%
ANIP	ANI Pharmaceuticals Inc Common Stock	$16,950	3.39%
SHLS	Shoals Technologies Group Inc	$10,300	2.06%
PHAT	Phathom Pharmaceuticals Inc	$8,650	1.73%
GLUE	Monte Rosa Therapeutics Inc	$15,250	3.05%
INOD	Innodata Inc	$16,850	3.37%
VERI	Veritone Inc	$15,950	3.19%
ALVO	Alvotech SA	$4,750	0.95%
ARRY	Array Technologies Inc	$17,450	3.49%

Stock Ticker	Company Name	Value ($)	% of total portfolio
FSLR	First Solar Inc	$9,300	1.86%
GEV	GE Vernova Inc	$1,250	0.25%
GILD	Gilead Sciences Inc	$1,700	0.34%
GL	Globe Life Inc	$1,600	0.32%
GLW	Corning Inc	$4,150	0.83%
GM	General Motors Co	$5,650	1.13%
GOOGL	Alphabet Inc Class A	$8,050	1.61%
GS	Goldman Sachs Group Inc	$3,400	0.68%
HCA	HCA Healthcare Inc	$2,900	0.58%
HIG	Hartford Insurance Group Inc	$650	0.13%
HII	Huntington Ingalls Industries Inc	$3,350	0.67%
HPQ	HP Inc	$15,950	3.19%
IDXX	IDEXX Laboratories Inc	$2,150	0.43%
IQV	Iqvia Holdings Inc	$1,800	0.36%
IVZ	Invesco Ltd	$6,600	1.32%
JBL	Jabil Inc	$2,200	0.44%
JNJ	Johnson & Johnson	$2,000	0.40%
JPM	JPMorgan Chase & Co	$1,900	0.38%
KLAC	KLA Corp	$6,100	1.22%
KO	Coca-Cola Co	$850	0.17%
KR	Kroger Co	$1,500	0.30%
L	Loews Corp	$2,350	0.47%
LEN	Lennar Corp Class A	$1,850	0.37%
LRCX	Lam Research Corp	$10,650	2.13%
LVS	Las Vegas Sands Corp.	$3,400	0.68%
MCK	McKesson Corp	$800	0.16%
MMM	3M Co	$1,000	0.20%
MNST	Monster Beverage Corp	$2,050	0.41%

Stock Ticker	Company Name	Value ($)	% of total portfolio
MO	Altria Group Inc	$2,950	0.59%
MPC	Marathon Petroleum Corp	$1,350	0.27%
MPWR	Monolithic Power Systems Inc	$6,750	1.35%
MRK	Merck & Co Inc	$2,200	0.44%
MU	Micron Technology Inc	$15,900	3.18%
NOC	Northrop Grumman Corp	$1,100	0.22%
NTAP	NetApp Inc	$2,700	0.54%
NTRS	Northern Trust Corp	$2,250	0.45%
PH	Parker-Hannifin Corp	$900	0.18%
PHM	Pultegroup Inc	$4,700	0.94%
RL	Ralph Lauren Corp	$3,300	0.66%
RTX	Rtx Corp	$1,350	0.27%
SOLV	Solventum Corp	$3,500	0.70%
STT	State Street Corp	$2,250	0.45%
STX	Seagate Technology Holdings PLC	$17,300	3.46%
T	AT&T Inc	$2,700	0.54%
TEL	TE Connectivity PLC	$2,800	0.56%
TGT	Target Corp	$1,000	0.20%
TMO	Thermo Fisher Scientific Inc	$550	0.11%
TPR	Tapestry Inc	$3,500	0.70%
TRV	Travelers Companies Inc	$550	0.11%
TXT	Textron Inc	$850	0.17%
UAL	United Airlines Holdings Inc	$4,000	0.80%
UHS	Universal Health Services Inc	$3,000	0.60%
VLO	Valero Energy Corp	$2,450	0.49%
WBD	Warner Bros Discovery Inc	$15,150	3.03%
WFC	Wells Fargo & Co	$1,350	0.27%
WRB	W R Berkley Corp	$1,000	0.20%



Total stock value in $: $415,400
Total stock % of Portfolio: 83%

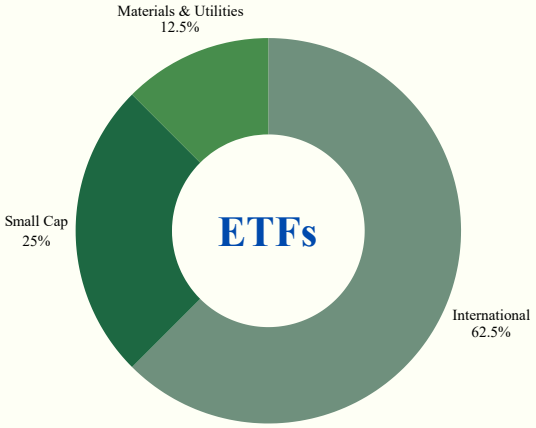


FINAL REPORT
Portfolio at a Glance 2

ETFs

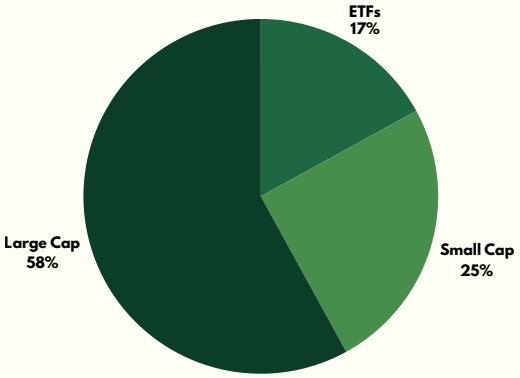
ETF Ticker	ETF Name	ETF # (from Approved List)	Value ($)	% of total portfolio
AVUV	AVUV AVANTIS US SMALL CAP VALUE	#71	$21,250	4.25%
XLB	MATERIALS SELECT SECTOR SPDR	#52	$8,400	1.68%
XLU	UTILITIES SELECT SECTOR SPDR FUND	#17	$8,400	1.68%
ESGE	ISHARES INC ISHARES ESG AWAR	#38	$14,100	2.82%
VSS	VANGUARD FTSE ALL-WORLD EX-US SMALL CAP ETF	#24	$12,800	2.56%
VEA	VANGUARD FTSE DEVELOPED ETF	#48	$19,650	3.93%

Total ETF value in $: $84,600
Total ETF % of Portfolio: 17%

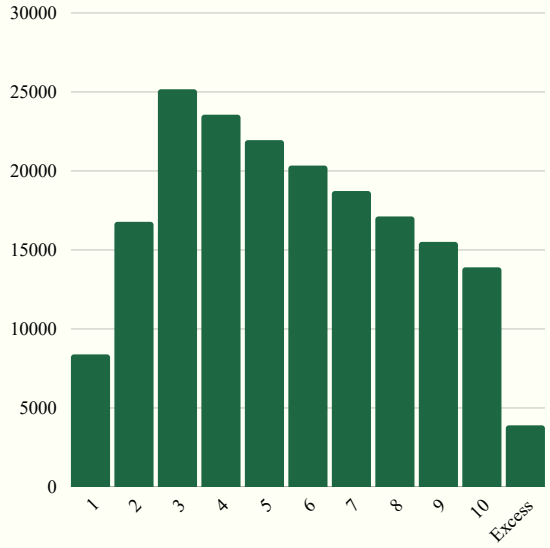


Cash

Total Cash Value ($): < $1.00
Total Cash % of Portfolio: < 1%



Projected Value of the Dedicated Annual Withdrawal Fund through the term (including annual withdrawals after year 3).



FINAL REPORT
Portfolio at a Glance 3

Our core belief to achieve Mr. Barwin's financial goals was to maximize growth potential. As a result, we took calculated risks driven by data and a need for diversification to maintain stability. 100% of Mr. Barwin's capital was invested in equities to ensure that we can achieve the strong return he requires in the long term. The core of the equities is the large-cap section, which ensures stability. To capture stock upside, a significant portion was allocated to more volatile small-cap companies. Finally, to provide the diversification and macro exposure, the remaining capital was allocated to ETFs.

Our portfolio was split into 58% large-cap stocks, 25% small-cap stocks, and 17% ETFs, a breakdown that provides the optimal balance of stability, high-growth opportunity, and broad diversification needed to meet Mr. Barwin's return goals.

Starting in the large-cap section of our portfolio, we diversified by ensuring adequate representation across each sector. We measured this by averaging the forecast from our Sector Rotation Prediction Model and the current sector distribution in the S&P 500. The result is a forward-looking, fairly representative sector distribution within our portfolio.

Limiting our small-cap holdings to ten stocks reflects both selectivity and high confidence. Our small-cap holdings support our team's thesis that emerging companies, particularly those in emerging sectors such as artificial intelligence, renewable energy, and biotechnology, offer asymmetric return potential.

The ETFs in our portfolio represent further diversification across international markets and underrepresented sectors such as materials and utilities. These holdings reinforce our recommended portfolio's stability without compromising its growth exposure.

Additionally, we believe that every holding in our recommended portfolio aligns with Mr. Barwin and the MTWB Foundation's core beliefs and values, whether that alignment was confirmed through our Scouting Report Index for large-cap stocks or through our qualitative analysis for small-cap stocks.

FINAL REPORT

Overview

Mr. Barwin presents the needs and desires of a successful, principled worker: modest short-term cash needs, combined with ambitious long-term goals. All of this is married to some core values that he requires us to adhere to. Our range of options, risk understanding, and flexibility allow us to tailor specific investment programs using existing, liquid investment alternatives.

To achieve Mr. Barwin's goals, we must:

- Generate a 12.3% compounded annual return to triple the principal by the end of the term.
- Fund annual payments of $10,000 starting after year three towards his foundation.
- Prioritize investments that match Mr. Barwin's values and the Make The World Better Foundation (MTWB) initiative.

To meet Mr. Barwin's objectives, particularly the significant growth required within his timeframe, is a demanding challenge, but we have a plan. By utilizing the laws of compounding, we can achieve the high returns needed to reach his target. Our portfolio will be allocated entirely to equities, allowing us to remove low-yielding assets that would slow growth without providing meaningful risk reduction. A 100% equity allocation enables us to focus on investments that are not excessively speculative yet have historically delivered returns exceeding the required benchmark. While an all-equity strategy may appear aggressive, each position is carefully selected and naturally hedged through diversification across sectors, markets, and market capitalizations. With disciplined execution, Mr. Barwin's capital can be positioned for maximum long-term growth. As team leader, Ben R. had known the other team members for several years and was familiar with their academic and personal interests and strengths. Therefore, the tasks and projects were allocated to suit those individual strengths. Ultimately, success comes down to building the right roster, so here is our starting lineup:

58% large-cap stocks, 25% small-cap stocks, and 17% exchange-traded funds.

The 58% large-cap stocks are our veteran players. These guys will follow and tend to meet or exceed the S&P 500's historical average 12% year-over-year return. We are confident in this because we have strategically selected the best of the best through a rigorous screening process. To begin, we looked at the S&P 500 and applied a proprietary quantitative index to immediately eliminate any stock that scored below the 70th percentile. Sam, our statistics expert, treats the S&P 500 like his fantasy football team: he built the index by comparing each company with peers in its industry and assigning a score based on its value and momentum.

Value was measured using companies' earnings yields, as this indicates which companies had the strongest earnings relative to their prices. Since raw metrics aren't comparable across sectors or companies, earnings yield was then converted into a standardized "z score." The "z score" tells us how many standard deviations above or below a stock is within its sector. If a stock had a negative z-score, it was more expensive than average. Conversely, a positive z-score indicates a stock is cheaper than average.

Momentum was evaluated using each company's 12-month price change, an industry-standard measure of trend strength. These raw momentum metrics were also converted to a "z score" measured in standard deviations away from the mean in a given sector. Each company's value and momentum z-scores were combined to form a composite score.

After eliminating the bottom 70th percentile in each sector, we were left with the strong companies that most closely represent the large-cap asset class. We also used each stock's index score to determine its weight within its sector.

Scouting Report Index

Mr. Barwin's goal is to grow capital in a way that aligns with his core values: supporting community-focused businesses with major local impact and a strong commitment to sustainability. Every investment in our portfolio needed to demonstrate these values in order to achieve his goal. The issue was that no standardized metric existed to assess companies on these qualitative measurements, and subjective judgment would inevitably lead to a mismatch with his stated goals.

Adam, our skilled coder, solved this by developing the Scouting Report Index (SRI), our proprietary solution for measuring how well a company matches with Mr. Barwin and the Make the World Better Foundation's mission. We made the SRI because it would be unwise to rely entirely on ESG grades and ESG indices to tailor our report to Mr. Barwin's goals. Typical ESG ratings were flawed due to their lack of standardized data and their inherent biases toward larger corporations. By avoiding the usual approach of filtering stocks based on ESG ratings, we created the SRI.

The SRI is generated through an automated process that reviews each company's 10-K filing and identifies targeted keywords associated with Mr. Barwin's values, such as "community," "sustainability", etc. After the initial extraction of the keyword, we calculated the frequency of these words by the total word count, as each 10-K filing is a different length. We then scored each result on a scale of 0 to 100. The higher the score, the more closely the corporation aligns with the foundation's values. Lower scores reveal a misalignment with Mr. Barwin's values, regardless of their financial strength.

Using this metric, we removed any company with a score below the 40th percentile in its sector. As a result, several high-performing large-cap names were excluded from our recommended portfolio due to misalignment with our value framework. One noteworthy case is Western Digital (WDC). It is ranked as our top quantitative performer in the Information Technology sector, with a proprietary index score of 26. Still, it received an SRI score of just 24.4, the fourth-lowest in its industry, leading to its disqualification. This disciplined process reveals that even if a stock appears financially compelling, it cannot enter the portfolio if it is not consistent with Mr. Barwin's values and MTWB's mission.

Sector Rotation Prediction Model

After completing the SRI analysis and confirming that each company in our large-cap universe aligns with the client's values, we then evaluated the portfolio's sector composition. Because we recognize the core principle of sector rotation, that different sectors outperform at various points in the economic cycle, we sought to build a systematic way to anticipate which sectors are likely to lead in the near future.

To accomplish this, we developed the Sector Return Prediction Model (SRPM), a machine learning model that forecasts 6-month forward returns for each sector. The SRPM incorporates 25 years of monthly macroeconomic and market data. It learns from indicators such as PMI Manufacturing, CPI YoY inflation, the 10yr–2yr yield-curve spread, unemployment rates, the Federal Funds Rate, and rolling sector returns/volatility. For each historical "economic moment," the model analyzes how those conditions correlated with sector performance six months later. Initially, we had planned to train our model to forecast 12 months in advance. However, our macroeconomic expert, Caleb, advised us that a 6-month forecast was more likely to be accurate than a 12-month forecast. After making that adjustment, our model demonstrated greater accuracy and confidence.

The model was trained on approximately 17 years of data and tested on the most recent eight years. Its out-of-sample R^2 values ranged from 0.68 to 0.74, meaning it explains roughly 68–74% of the variation in future 6-month sector returns—strong predictive power for a financial forecasting model.

Under current macroeconomic conditions, the SRPM outputs the following expected 6-month returns for our investable sectors:

- Consumer Staples: +1.0%
- Healthcare: +1.6%
- Financials: +1.6%
- Technology: +7.9%
- Energy: +1.4%
- Communication Services: +2.1%
- Industrials: +2.8%
- Consumer Discretionary: +2.0%

To convert these return forecasts into actionable weights, we first exclude sectors with negative expected returns. We then normalize each sector's expected return by dividing it by the sum of all positive expected returns, yielding a set of proportional weights that sum to 100%. This gives us a predictive, forward-looking sector allocation.

Finally, we blend these model-derived weights with the current sector distribution of the S&P 500 to maintain while still capturing the model's signals. We rebalance every six months, matching the model's forecasting horizon, to ensure our portfolio stays aligned with the sectors most likely to outperform in the upcoming cycle.

Small-Cap

Our 25% small-cap allocation serves as the foundation for achieving Mr. Barwin's long-term growth goal: established large-cap companies, by nature, often cannot deliver. Given the complexity of the small-cap space, we developed a systematic, quantitative, and qualitative process to evaluate and select small-cap companies. Our philosophy here is simple: we are not interested in speculation, but rather evidence. The two questions we consistently asked were:

1. *Is the company growing financially?*
2. *Is the company operating in a market with the upside to support long-term, sustainable growth?*

These questions led us to develop our Small Cap Quantitative index, a quantitative scoring metric tailored to small-cap financials. Each company is assigned a rating based on its performance across the following metrics: Operating Cash Flow, Revenue Growth, and Price-to-Book Ratio. Operating Cash Flow tells us whether the company's core operations generate or consume cash. It highlights the difference between healthy reinvestment and unsustainable burnout by assessing whether current operations can scale sustainably without constant funding. Quarterly Revenue Growth measures how quickly the market is adapting to the company's product or technology. Strong YoY growth indicates high customer demand and signals that the company's idea has moved from concept to product. The Price-to-Book Ratio helps us assess a company's valuation relative to its assets. A lower PB indicates that the market has not fully

priced in future growth, while a higher PB tends to signal major innovation and/or revenue growth. Once a company passes our quantitative screening, it proceeds to the qualitative stage. Here, we analyze leadership capability, product, competition, and scalability. This stage required the most collaboration within our team. Noah, our research expert, assembled the team and provided a detailed agenda for the companies we needed to decide on. As we debated each company's strengths and risks, we ensured that no decision was made without strong evidence. Our qualitative strategy was centered on the Total Addressable Market (TAM) Analysis, which shows that a company must be positioned in a large, expanding market to sustain multi-year growth.

Like promising young players entering the league, these higher-risk companies require careful oversight throughout the term, and we will be ready to make strategic adjustments and execute trades whenever necessary to keep the portfolio protected and performing at its highest level.

Exchange-Traded Funds

In our recommended strategy, the 17% allocated to ETFs serves as a diversification tool. Victoria, our team's most reliable stabilizer, knew precisely how our portfolio could benefit from diversification. These ETFs allow the portfolio to gain exposure to international markets, small-cap companies, and sectors like materials and utilities, which require a large number of holdings for proper diversification. Our international stock-based ETFs serve as diversifiers that offer the same competitive growth potential as our core U.S. large-cap holdings but can provide some safety if the domestic market stumbles. Exposure to small-cap companies as a whole allows us to benefit from post-recession markets, when small caps tend to thrive. The ETFs we are proposing also have high yields, averaging 2.3%, which contribute to our total portfolio dividend yield needed for Mr. Barwin's annual withdrawals. Think of ETFs as the kicker who wins the game with a 58-yard field goal and then disappears into the tunnel before interviews.

Long Term Plans

A glide-path strategy gradually reduces a portfolio's risk as the investor nears the point when the funds will be needed. Without it, an investor could face significant late-stage volatility, risking a major loss right before withdrawal, potentially undoing years of progress. However, a linear glide path strategy can also be detrimental if most of your gains only come towards the end of the term. For this reason, we will utilize a Dynamic Asset Allocation (DAA) system. Beginning in year 8, the asset allocation of our portfolio will be determined by the DAA formula:

$$\text{Equity Weight}(t) = f(P_{\text{success}},\ \text{shortfall risk},\ \text{return gap})$$

This approach continually adjusts the portfolio based on changes in the probability of success, shortfall risk, and the return still required to reach our goal – adding risk when necessary to stay on track or reducing risk to preserve gains. By doing so, we ensure the portfolio remains aligned with the most efficient path to achieving Mr. Barwin's objective. We will also rebalance the entire portfolio regularly, ensuring that if one asset class declines, the strength and stability of our other holdings provide the buying power needed to strategically purchase undervalued positions and keep the portfolio advancing toward its goal.

We're securing the ball, we're not fumbling, and we're getting Mr. Barwin to the finish line.

Together, we enhanced our creativity and strategy by keeping our eyes on the endzone and thinking from our client's perspective. Furthermore, clear communication and problem-solving ensured we moved the chains together and continued to gain first downs. As a team, we recognized that each person brought expertise we needed to tap into. This eventually enabled us to come together and discuss how to leverage each other's strengths and build on the previous team members' research to collectively excel. We understood it would not be easy; however, we persisted, pushing each other and embracing the true meaning of camaraderie and teamwork to carry Mr. Barwin to the endzone.

Work Cited

ChatGPT. (2025). *Assistance in developing Scouting Report Index (SRI) model and quantitative framework* [Large language model]. OpenAI.

Google Colaboratory. (2025). *Python notebook environment used for machine-learning development*. Google.

Hastie, T., Tibshirani, R., R., & Friedman, J. (2009). *The elements of statistical learning: Data mining, inference, and prediction* (2nd ed.). Springer.

Investopedia. (2024). *Momentum investing*. investing

Investopedia. (2024). *Earnings yield* (E/P ratio).

J.P. Morgan Asset Management. (2024). *Guide to the markets: U.S. equity sector performance and macroeconomic drivers*.

McKinsey & Company. (2023). *The future of AI, biotechnology, and clean energy: Global growth projections for emerging innovation sectors*.

Russell, A. (2023). *How to implement Random Forest from scratch with Python* [Video]. YouTube.

Standard & Poor's. (2024). *S&P 500 sector breakdown and methodology*.

Vanguard. (2024). *Vanguard FTSE Developed Markets ETF* (VEA): Performance & yield data.

Vanguard. (2024). *Vanguard FTSE All-World ex-US Small-Cap ETF* (VSS): Performance & yield data.

Avantis Investments. (2024). *Avantis U.S. Small Cap Value ETF* (AVUV): Performance overview.

State Street Global Advisors. (2024). *The Energy Select Sector SPDR Fund* (XLE).

FT Portfolios. (2024). *First Trust Cloud Computing ETF* (SKYY).

Global X ETFs. (2024). *AIQ: Artificial Intelligence & Technology ETF*.

U.S. Bureau of Labor Statistics. (2024). *Consumer Price Index (*CPI).

Federal Reserve Bank of St. Louis. (2024). *Unemployment rate, yield curve data, and Fed funds rate* (FRED database).

Institute for Supply Management. (2024). *Purchasing Managers Index* (PMI®) reports.